

Response to the Associate Editor

“The Hedged Random Forest”

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E. Beck

D. Kozbur

M. Wolf

We thank the Associate Editor for the encouraging assessment of the paper’s potential and for the constructive guidance on how to strengthen it. Below we address each of the three points in turn. All section, table, and figure references are to the revised manuscript.

Comment 1: A financial-econometric application

The empirical section would benefit from a closer match with the readership and standards of Journal of Financial Econometrics. The current evidence is useful in showing that the proposed hedged random forest improves upon the standard random forest and several weighted-random-forest variants on benchmark regression datasets. However, it does not yet fully demonstrate how the method performs in a financial-econometric setting. I would encourage the authors to add one substantive financial-econometric application, such as return prediction, volatility or realized-volatility forecasting, option-implied volatility forecasting, credit-risk prediction, macro-financial forecasting, or risk-premium estimation. The application should be framed as a genuine out-of-sample forecasting exercise, with rolling or expanding-window evaluation where appropriate. At the same time, I do not think the paper needs to become a broad horse-race comparison against all modern tree-based algorithms.

Response. We have added a new application, Section 4.5, that directly addresses this comment. Rather than a single hand-picked series, we forecast a broad panel of 41 target series drawn from two groups of the FRED-MD database [McCracken and Ng, 2016]: interest-rate levels, credit and term spreads, and exchange rates (the *Interest and Exchange Rates* group), and price indices (the *Prices* group). Scanning across an entire class of targets, rather than reporting a single series, was a deliberate choice: it lets us show *where* the hedging mechanism helps within a coherent financial-econometric setting, without turning the exercise into a horse-race across competing algorithms, which you correctly note the paper does not need.

The exercise is a genuine expanding out-of-sample forecasting exercise in the sense you describe: for each target and each of 432 monthly origins (1990-05 through 2026-04), we fit the random forest (and HRF) on a rolling 360-month window, forecast the next month, and record the realized forecast error. We adapt the time-series estimators of $\hat{\mu}$ and $\hat{\Sigma}$ developed

in our companion paper on inflation forecasting [Beck and Wolf, 2025] (EWMA estimation combined with linear shrinkage) rather than the off-the-shelf estimators used for the i.i.d. benchmark datasets in Section 4.2, since — as we discuss in Section 4.5 — suitable estimators of the two hedging inputs depend on the nature and frequency of the data for time-series applications.

Table 3 and Figure 4 summarize the results by group, with the full 41-series breakdown in Appendix B, Table 13. Across all 41 series, HRF achieves a lower RMSE than RF for 32 series (78%), with a mean RMSE ratio of 0.982. The gains are strongest and most uniform for the eight credit and term spreads (HRF wins for every single series, mean ratio 0.938), more modest for the price indices (mean ratio 0.991), mixed for interest-rate levels (mean ratio 0.995, concentrated at the short end of the curve), and essentially absent for exchange rates (mean ratio 1.000), consistent with the well-documented difficulty of beating a random walk in exchange-rate forecasting. This last point is itself informative: it shows that HRF does not manufacture spurious gains where none should be expected.

Comment 2: Incremental value relative to modern tree-based methods

Refining the contribution is important because unequal and signed contributions of trees are already familiar in boosted-tree methods. The authors should therefore show whether HRF offers incremental value relative to modern tree-based machine-learning tools, or clarify that its contribution is instead a low-tuning post-estimation improvement to random forests. A carefully designed financial-econometric application, together with some diagnostic evidence on when the weighting scheme works, would make the empirical contribution considerably more convincing.

Response. We take the second of the two options you outline: we clarify, rather than attempt to out-perform, boosted-tree methods, for the following reason. Gradient boosting [Friedman, 2001] and its variants construct an entirely different ensemble than the random forest: trees are fit sequentially to the residuals of the current ensemble, and performance typically depends on the joint tuning of several hyperparameters (the learning rate, tree depth, number of boosting rounds, and regularization). HRF, by contrast, is a *post-estimation* re-weighting of an already-trained, standard random forest. It requires no change to how the individual trees are grown and no retraining, and it has a single user-facing hyperparameter, the leverage constraint κ , for which we recommend — and use throughout the paper, including in the new Section 4.5 — the default choice $\kappa = 2$ (see Section 3). We therefore view HRF as a low-tuning, “point-and-shoot” improvement (this is literally how we describe the i.i.d. estimators in Section 4.5) that a random-forest user can apply on top of an existing forest, rather than as a competitor to boosting-type ensembles built from the ground up. We agree that a direct horse-race against boosting would be a natural and interesting extension, but, per your own comment above, we have deliberately kept the paper’s empirical scope to HRF versus RF and previously proposed weighted random forests, rather than broadening it into a comparison against all modern tree-based algorithms. This is a deliberate scoping strategy

on our part, not an oversight: HRF is positioned throughout the paper as a post-estimation refinement of a standard random forest, and we view a fair, properly tuned comparison against boosted-tree methods as a separate undertaking that deserves its own paper rather than a section here.

TODO (internal — remove before submission): We are about to tell the AE that we will *not* run a head-to-head comparison against boosting or other modern tree ensembles in this paper — framing HRF purely as a post-estimation add-on and pushing a boosting comparison to future work. This is a real strategic commitment, and it isn't settled on our end yet: we should all agree before this goes out. If we'd rather leave the door open (e.g. a referee could ask for this in a later round, or we may want it for a follow-up paper), let's discuss before submitting.

On the diagnostic evidence you request: the FRED-MD results just described already provide one clear pattern. The gains from HRF are largest and most uniform for the credit and term spreads, a group in which the underlying predictors (default risk, liquidity conditions, and the level of rates) are heterogeneous and only loosely related, plausibly giving rise to correspondingly heterogeneous, less mutually redundant tree forecasts for HRF to exploit; the gains are essentially absent for exchange rates, which are close to unpredictable by any method. A second, independent piece of diagnostic evidence was reported in our response to Reviewer 1 (Additional Comment 1): across the 17 i.i.d. benchmark datasets of Section 4.2, the dispersion in individual trees' out-of-bag accuracy is the one dataset characteristic (of five considered) that is significantly related to HRF's performance gain (Spearman $\rho = -0.72$, $p = 0.002$), with datasets with the most heterogeneous trees showing the largest gains. Both pieces of evidence point in the same direction: HRF helps most when trees are not all equally reliable, which is exactly the situation its weighting scheme is designed to exploit.

Comment 3: Similarity score and prior circulation

Lastly, the similarity check reports a similarity score of 42%, which has raised some concern at the editorial office. Having reviewed the report, I understand that much of the overlap appears to be with an earlier working-paper version of the same project. The authors would benefit from further revising the writing, presentation, and framing of the manuscript so that the submitted version is more clearly differentiated from the earlier version, while ensuring that any prior circulation is properly acknowledged.

Response. You are correct: an earlier version of this paper circulated under the same title as a working paper, Beck et al. [2024]. This JFEC submission is the first formal journal submission of that project; the working paper itself was never submitted elsewhere. We had not yet added an explicit acknowledgment of the prior circulation to the manuscript and are glad to add a title-page footnote citing the working paper if the editorial office considers this necessary — we did not want to make further changes to the main text before confirming the exact wording and placement you and the editorial office would prefer.

On differentiation from the working paper: relative to that version, the manuscript has been substantially revised, both in response to Reviewer 1 and in response to this letter.

Reviewer 1’s comments led us to add two additional benchmark comparisons not present in the working paper — a ridge combination scheme and a bias-free minimum-variance variant, HRF-MinVar (Section 4.4.1) — a discussion of negative weights in relation to tree correlations (Section 4.3), a remark clarifying that the leverage constraint yields dense rather than sparse solutions (Remark 3.4), and a unification of notation throughout the paper around the shrinkage/sample $\times \kappa \in \{1, 2\}$ labeling scheme. This letter adds a further, substantial change: the new financial-econometric application of Section 4.5, which has no counterpart in the working paper. Between the two rounds of revision, a large share of the paper’s empirical content and exposition is new relative to the circulated working paper, and we would welcome any further guidance from the editorial office on presentation changes that would help differentiate the two versions.

References

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- Jerome H Friedman. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232, 2001.
- Michael W McCracken and Serena Ng. FRED-MD: A monthly database for macroeconomic research. *Journal of Business & Economic Statistics*, 34(4):574–589, 2016.